

San Francisco Condo Market Analysis

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Summary

This analysis illustrates key factors influencing property prices across different districts. One prominent insight is the significant price premium associated with District 7, where homes command nearly \$387,000 more than average, driven by location-based desirability. This effect is mirrored in the **Square Footage Value by District** plot, which highlights the dramatic disparity in price per square foot—exceeding \$1,000/sqft in District 7 but falling below \$200/sqft in District 10, reflecting socioeconomic and infrastructural variations.

Additionally, the **Square Footage Effect Across Price Tiers** visualization emphasizes how the impact of square footage intensifies in luxury markets, where homes in the top 90th percentile show a 2.5× greater return on added square footage than mid-tier properties, indicating that size matters more at higher price points. Interestingly, the **Bedroom Effect Across Price Tiers** chart reveals a counterintuitive trend: increasing the number of bedrooms may actually reduce property value when controlling for square footage, suggesting that buyers prioritize fewer, more spacious rooms.

Furthermore, the **Days on Market (DOM)** and **Condo Sales by Month** plots offer insights into market dynamics and liquidity, illustrating seasonal variations and temporal trends. Despite these fluctuations, the analysis finds no consistent relationship between condo sales volume and changes in the Federal Reserve interest rates, reflecting the complex interplay of local housing demand and broader macroeconomic forces.

Collectively, these visualizations provide a nuanced, data-driven understanding of San Francisco's real estate market, underscoring the multidimensional factors that shape property valuation and buyer behavior.

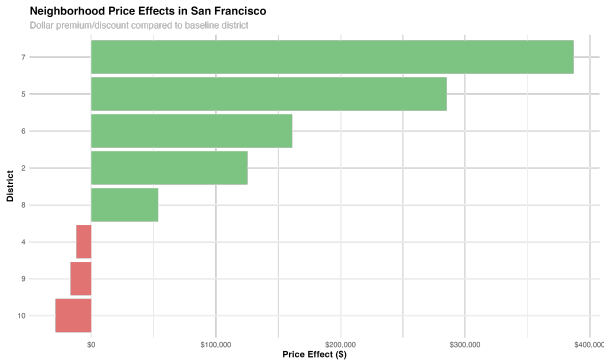
Key Insights

- Location is the dominant factor in property pricing, with District 7 commanding a premium of nearly \$387,000
- Square footage value varies dramatically by location - worth over \$1,000/sqft in District 7 vs. under \$200/sqft in District 10
- Impact of square footage increases with property price tier - 2.5× greater in luxury properties (90th percentile)
- Additional bedrooms decrease property value when controlling for square footage, suggesting buyers prefer fewer, larger rooms
- Property age effects vary by price segment, with mixed evidence about whether newer properties command higher prices
- There is no significant relationship between the number of condo sales per month and fluctuations in the Federal Reserve's interest rates

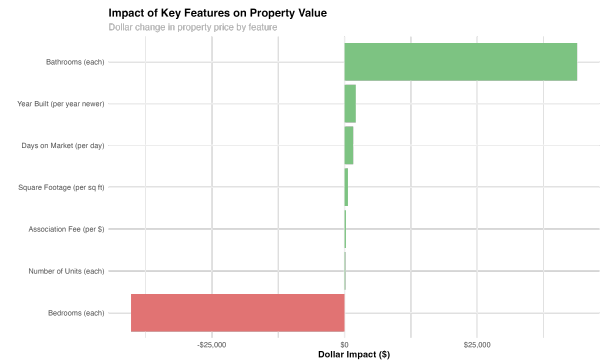
Tools

- R libraries dplyr, ggplot2, tidyr, scales, lubridate, quantmod, randomForest, ggpubr, lme4, mgcv, quantreg, glmnet, sjPlot, gratia, and vip

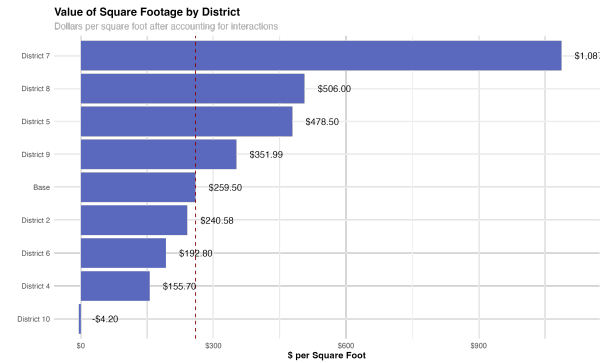
Neighborhood Price Effects



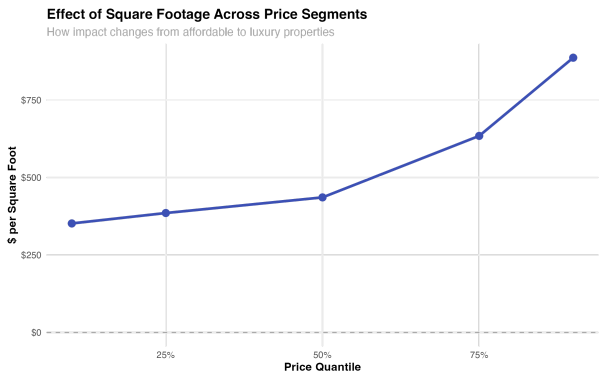
Feature Importance



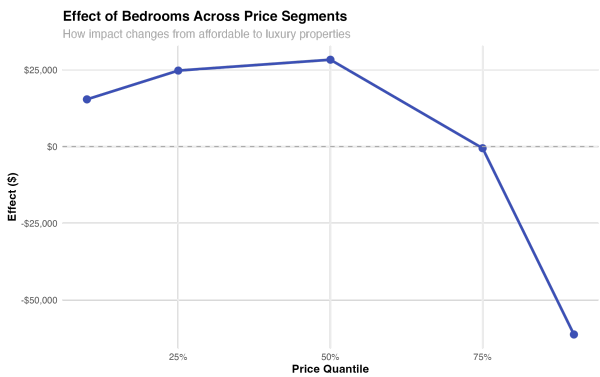
Square Footage Value by District



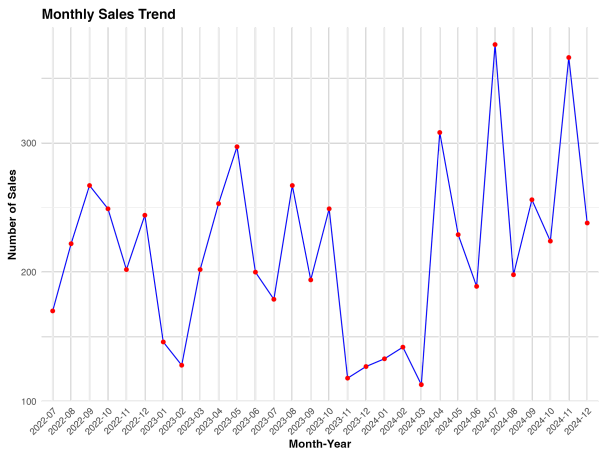
Square Footage Effect Across Price Tiers



Bedroom Effect Across Price Tiers



Condo Sales by Month



Distribution of Days on Market (DOM) Over Time

The graph illustrates the trend of the average number of days a property remains on the market (DOM) from mid-2002 to early 2005. The y-axis represents the DOM in days, ranging from 90 to 130. The x-axis represents time in month-year format, with labels every three months. A solid red line indicates the mean DOM, while the gray shaded area represents the distribution of DOM values. The mean DOM starts at approximately 90 in 2002-06, rises to a peak of about 130 in 2003-05, then declines to around 108 in 2004-09, and finally rises slightly to 110 in 2005-01.

Time (Month-Year)	Mean DOM (Days)	Lower Bound (Days)	Upper Bound (Days)
2002-06	90	85	95
2002-09	100	95	105
2002-12	110	105	115
2003-03	120	115	125
2003-06	128	123	133
2003-09	120	115	125
2003-12	118	113	123
2004-03	115	110	120
2004-06	110	105	115
2004-09	108	103	113
2004-12	108	103	113
2005-01	110	105	115

The chart displays the monthly average adjusted selling price for a property. The y-axis represents the price in dollars, ranging from \$400,000 to \$1,000,000. The x-axis shows the months from July 2022 to December 2024. The price shows significant volatility, with a major peak in early 2023 and a general downward trend towards the end of the period.

Month-Year	Average Adjusted Price (\$)
2022-07	650000
2022-08	430000
2022-09	600000
2022-10	520000
2022-11	420000
2022-12	390000
2023-01	700000
2023-02	980000
2023-03	550000
2023-04	630000
2023-05	580000
2023-06	500000
2023-07	580000
2023-08	500000
2023-09	1050000
2023-10	610000
2023-11	560000
2023-12	430000
2024-01	610000
2024-02	540000
2024-03	470000
2024-04	590000
2024-05	470000
2024-06	580000
2024-07	630000
2024-08	620000
2024-09	480000
2024-10	500000
2024-11	430000
2024-12	460000

A histogram titled "Distribution of Square Footage" showing the frequency of square footage values. The x-axis is labeled "Square Footage" and ranges from 0 to 5000. The y-axis is labeled "Count" and ranges from 0 to 600. The distribution is unimodal and right-skewed, with a peak count of approximately 700 for square footage between 500 and 750. The data is represented by orange bars.

Fed Funds Rate vs Number of Sales

Correlation: $r = -0.046$

$R^2 = 0.002$
p-value = 0.809